

Reaction Rates and Rate Laws

Rates, rate laws, units of k and first-order half-life

Name _____ Class _____ Date _____

USEFUL RELATIONSHIPS

$$\text{rate} = -\frac{\Delta[\text{reactant}]}{\Delta t} \cdot \text{rate} = k[A]^m[B]^n \quad \text{First order: } [A]_t = [A]_0 e^{-kt} \cdot t_{1/2} = \frac{0.693}{k}$$

Give the units of every rate and every rate constant.

A. Understanding

- Use collision theory to explain why (a) powdered marble reacts faster with acid than a single lump of the same mass, and (b) the rate increases with concentration.

B. Rates and stoichiometry

- In $2\text{N}_2\text{O}_5(\text{g}) \longrightarrow 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$, N_2O_5 is used up at 0.0060 M/s. Find the rates at which NO_2 and O_2 form.

- The concentration of a reactant falls from 0.800 M to 0.560 M in 120 s. Calculate the average rate of its disappearance.

C. Rate laws

4. The rate law for $2\text{NO} + 2\text{H}_2 \longrightarrow \text{N}_2 + 2\text{H}_2\text{O}$ is $\text{rate} = k[\text{NO}]^2[\text{H}_2]$. State (a) the order with respect to each reactant (b) the overall order (c) the units of k (d) the effect on the rate of doubling $[\text{NO}]$.

5. For $2\text{NO} + \text{O}_2 \longrightarrow 2\text{NO}_2$: Experiment 1: $[\text{NO}] = 0.020\text{ M}$, $[\text{O}_2] = 0.010\text{ M}$, $\text{rate} = 2.8 \times 10^{-6}\text{ M/s}$. Experiment 2: $[\text{NO}] = 0.040\text{ M}$, $[\text{O}_2] = 0.010\text{ M}$, $\text{rate} = 1.12 \times 10^{-5}\text{ M/s}$. Experiment 3: $[\text{NO}] = 0.020\text{ M}$, $[\text{O}_2] = 0.020\text{ M}$, $\text{rate} = 5.6 \times 10^{-6}\text{ M/s}$. Determine the rate law and the value and units of k .

6. Using your rate law from question 5, what happens to the rate if $[\text{NO}]$ is doubled and $[\text{O}_2]$ is halved at the same time?

7. Give the units of k for a zero-order, a first-order and a second-order reaction (concentrations in M , time in s).

D. Half-life

8. A first-order reaction has $k = 2.31 \times 10^{-3}\text{ s}^{-1}$. (a) Find its half-life. (b) If $[\text{A}]_0 = 0.400\text{ M}$, what is $[\text{A}]$ after 900. s?

E. Challenge

9. A first-order reaction has $k = 0.0150 \text{ min}^{-1}$ and $[A]_0 = 0.200 \text{ M}$. Calculate (a) $[A]$ after 60.0 min and (b) the time for $[A]$ to fall to 0.0500 M.